



LEEDS CITY COUNCIL

Highways and Transportation

Highways Services
Civic Hall
Calverley Street
Leeds LS1 1UR



Junction: A660 Headingley Lane, near Richmond Avenue

Company: Telent

Controller Type: Optima

Mains Supply: 230V / 50Hz

New / ~~Modification~~

Operating Voltage: ELV

Engineer: Reza Yazdanimotlagh

Telephone No: 0113 3367359

OTU SCN 03440

Cont SCN 03441

IP Address Information:

Controller IP:

Gateway:

Mask:

OTU IP:

NOTE TO UTC PERSONNEL

This specification should be used for general purposes only.

The Signal Company documentation should be used for controller operation and timings unless amended on this sheet as RAM data.

RAM UPDATE (If no data then write NONE)	Reasons for change	Initials	Date
Write date current EPROM fitted:-			

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378L A660 Headingley Lane, near Richmond Avenue

Configuration Notes

General

A Telent 4-phase Optima ELV controller from contract 3664 shall provide the method of control in this specification.

The site is a Toucan crossing with kerbside and on-crossing detection.

Vehicle detection is with cut carriageway loops.

The normal method of control shall be MOVA.

Traficones to be addressed through 1 BPL2 card with IP:

VA Pre-timed Max with additional max timer

There will be TWO max green timers. The first will be pre-timed. The 2nd VA max timer (DA), set to 5 seconds will start upon receipt of a ped demand. The pre timed max will commence after the start of vehicle green and will run for 20 seconds. If during this time a pedestrian demand is registered then gap change will be permitted, should no vehicle extensions be present. After the pre timed max has expired and there has been no pedestrian demand the second VA max timer will govern operation. If a demand for the pedestrian is received during the pre-timed max period and there is still some residue of the second VA max then the second max will govern operation and continue to extend the traffic phase. If the second VA max has expired at the end of the pre-timed max then the pedestrian demand will be activated.

Green Man Extensions.

It shall be possible to extend the green man period when operating under VA and MOVA control. Maxsets values have been assigned to the pedestrian phases and the associated kerbside detectors conditioned to extend their phase.

Kerbside Detector Operation.

The kerbside detectors will operate in 3 ways:

1. To eliminate the requirement for pedestrians to place a demand using the push button the kerbside detectors shall place an unlatched demand for pedestrian phase C, initially configured with a call delay of 2 seconds and a cancel period of 2 seconds (alterable by handset). This operation shall be configured with a software switch that can be switched by timetable and initially set to active.

2. The general operation of the kerbside and pushbutton demands will be as set out in TOPAS 2500. Pushbutton demands will be held and cancelled depending on the active state of the kerbside and any push button demand placed when no kerbside presence is detected shall place a latched demand.

3. The raw output from the kerbside detector shall set MOVA Det 41. This shall be used to extend the green man period through the use of priority links within MOVA.

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Audible Signal

An audible signal shall be present during the green man period with the facility to be switched by timetable. Initially set to turn off the audible signal everyday between 22:00 – 07:00.

Mova Conditioning

Pre-timed Max in MOVA 8:

Mova operation will be similar to VA pre-timed max operation. Changes to timings will be implemented using MOVA 8 special conditioning. The timings will be the same as VA sites with a 20 second delay timer and an additional 5 second timer. VA pre-timed max still to be conditioned in the controller at MOVA sites in case of MOVA failure.

Operation using MOVA special conditioning

A delay timer starts (20 seconds) at the start of vehicle green. After this has expired a variable is set that is only reset when the pedestrian demand is serviced. If a pedestrian demand is requested once the variable has been set then it starts an additional 5 second timer, after which the pedestrian demand would be forced by a priority link demand. This priority demand can be inhibited by a bus priority request.

MOVA UTC Interface:

Control: F1 F2 MTO DS1 DS2 DS3 DS4

Reply: G1 G2 CRB MO

Wait Lamp Mimic: Whilst pedestrian phase wait lamp is lit, input to MOVA:

Ped C = MOVA Det 9

CRB Toggle: CRB configured to toggle every 120 seconds if MOVA is not in control.

UTC & MOVA confirmation LED

Within the manual panel the UTC control **confirmation LED shall FLASH when the controller is under MOVA control and remain on solid when under UTC control.**

A clearly labelled, manually selected VA switch shall be provided within the manual panel which shall override MOVA control when active.

MOVA Loop Positions

Detectors shall be wired into packs as shown below, to prevent chattering.

To assist in validation all detector inputs to be clearly labelled on detector packs in controller.

Approach	Loop	Dist From Stop Line (m)	Longitudinal Dimension (m)	Shape	Clearance Distance		Detector Pack No.
					N/S	O/S	
A660 Headingley Lane Inbound	AIN1	80	1.7	Diamond	0.4	0.8	1
	AX2	40	1.7	Diamond	0.4	0.8	1
A660 Headingley Lane Outbound	BIN3	85	1.7	Diamond	0.4	0.8	1
	BX4	40	1.7	Diamond(Uni-Directional)	0.4	0.8	1

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Flow & Occupancy Loops

To monitor the various profiles of vehicles passing through the junction the following conditioning is required on each of the detection loops listed below: -

FLOW - After every 4 vehicles that pass over AIN1, BIN3 change the state of the reply bit.

OCCUPANCY – Assigned IN or X loop reply shall mimic the raw loop detector output.

Loop	Reply bits (Flow/Occ/Queue)	TMS SCN No.
AIN1	21/25	SCN 43858
BIN3	22/26	SCN 43859

Corridor Control Strategy

The following control bits are processed by UTC and sent based on different traffic conditions on the corridor. The purpose of these is to hold vehicle green to flush traffic through when required.

If control bit 17 (SF1) is received, pulse MOVA Det 29 for 2 seconds. (SCN 43854)

If control bit 18 (SF2) is received, pulse MOVA Det 30 for 2 seconds. (SCN 43855)

If control bit 19 (SF3) is received, pulse MOVA Det 31 for 2 seconds. (SCN 43856)

If control bit 20 (SF4) is received, pulse MOVA Det 32 for 2 seconds. (SCN 43857)

Dynamic Dataset Changes - Based on the flow/occupancy/queueing conditions of the corridor it is desirable to change MOVA Datasets. The following logic and MOVA detectors facilitate the changes.

If control bit 12 (DS1) is being received from UTC, output to MOVA Det.25 (SCN 43850)

If control bit 13 (DS2) is being received from UTC, output to MOVA Det.26 (SCN 43851)

If control bit 14 (DS3) is being received from UTC, output to MOVA Det.27 (SCN 43852)

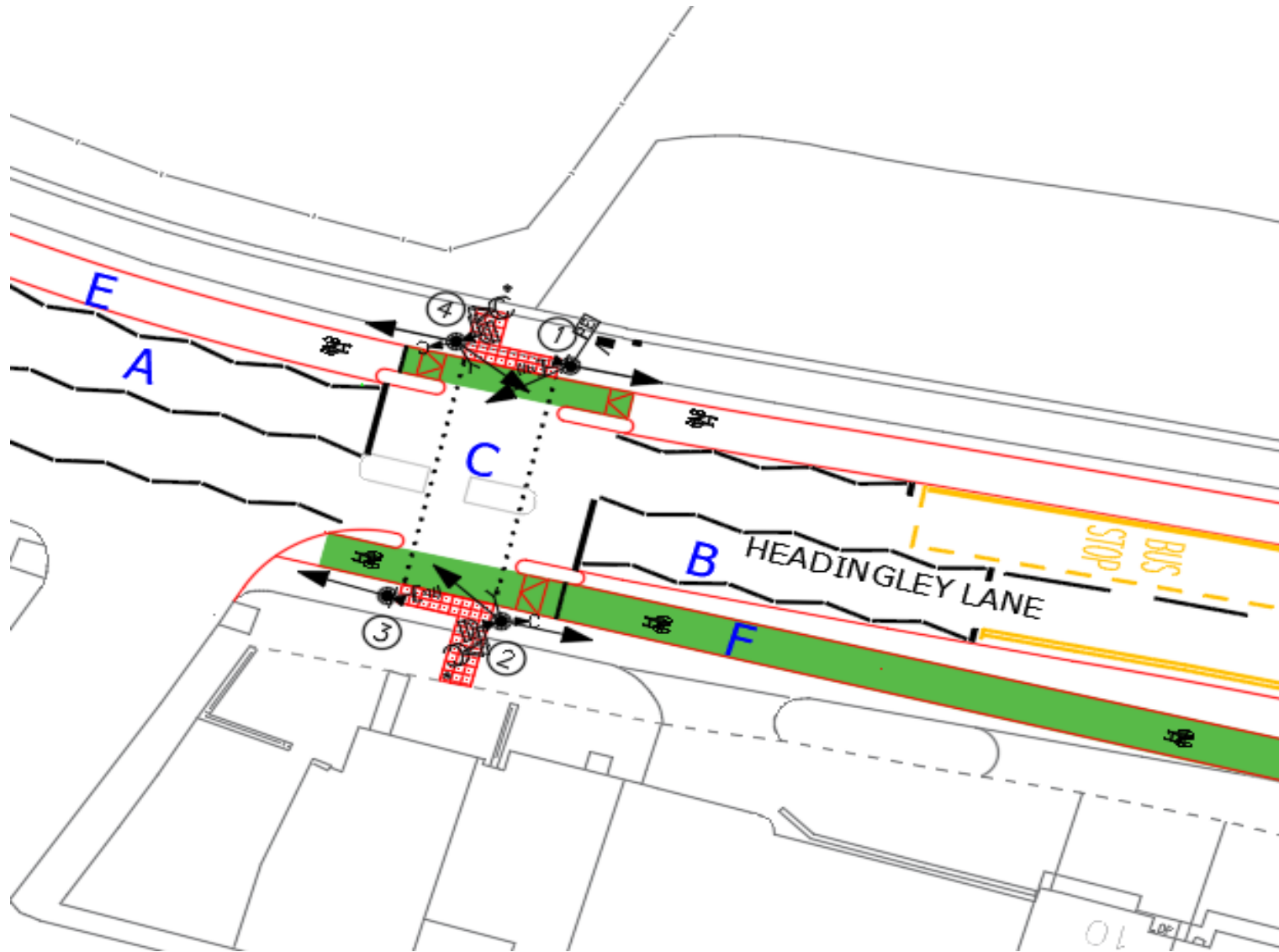
If control bit 15 (DS4) is being received from UTC, output to MOVA Det.28 (SCN 43853)

Bus Priority

Implemented via RTIG Bus Triggers shall be configured and input to MOVA dets 33 through to 38 respectively.

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Site Layout



NOT TO SCALE

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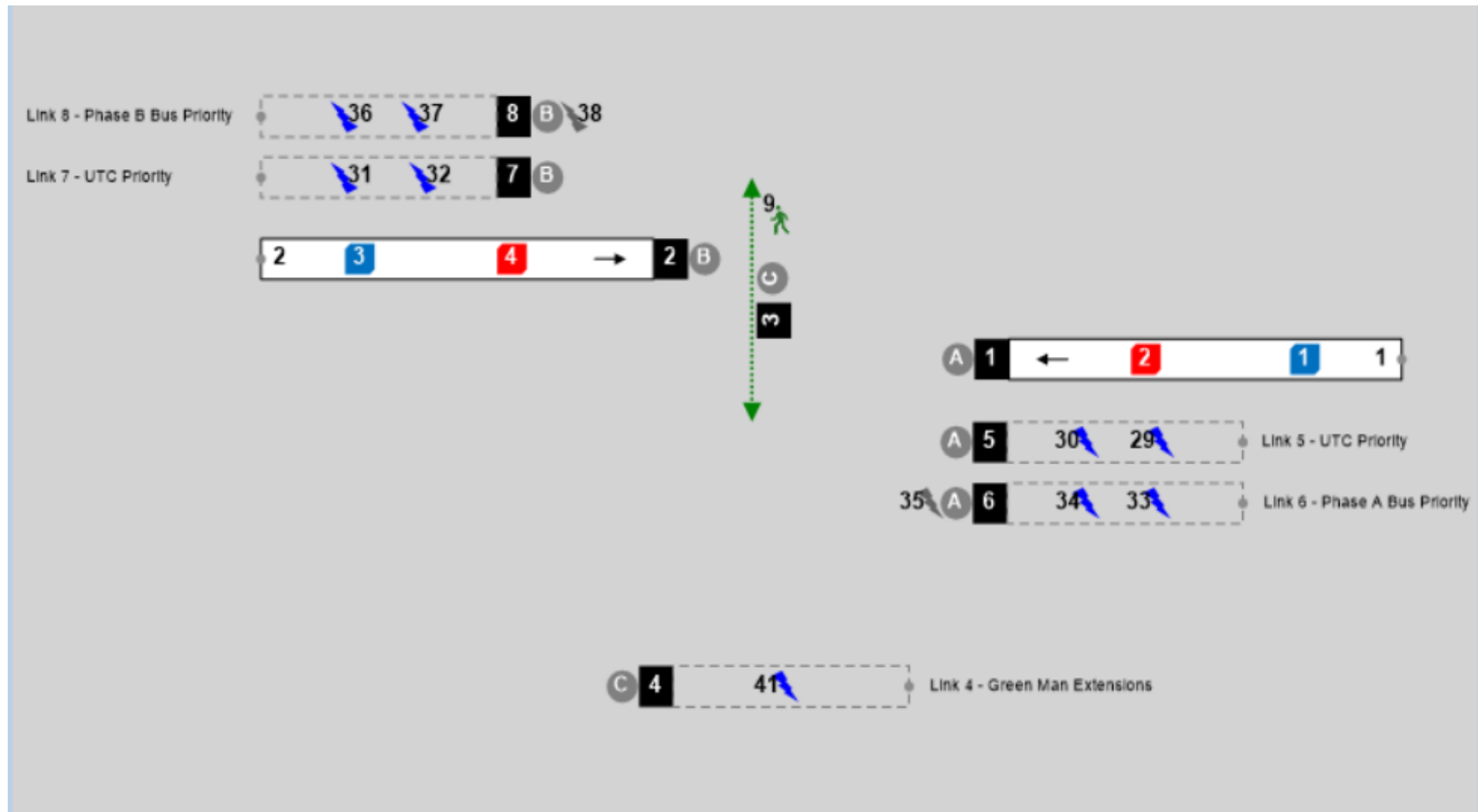
Site Layout



NOT TO SCALE

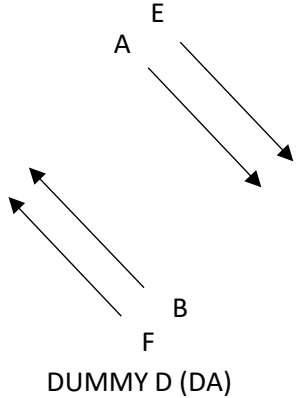
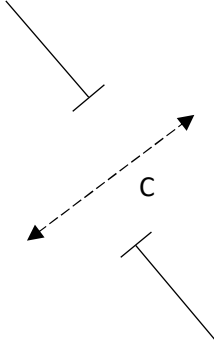
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MOVA Link Diagram



NOT TO SCALE

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BASIC STAGE DATA FOR STREAM NO 1			
Stage: 0	Stage: 1	Stage: 2	Stage: 3 Remove
MANUAL ALL RED			
Stage:	Stage:	Stage:	Stage:

	STREAM 1	STREAM 2	STREAM 3	STREAM 4
ALL RED STAGE	0			
START UP STAGE	1			
REVERSION STAGE	1			

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[illegible]

Note 1:- Condition of Appearance

0 - Phase always appears when stage runs

1 - Phase only appears if demand exists at start of inter-stage

2 - Phase appears, if demanded, at any time up until the end of the stage in which the phase runs

3 - Phase appears, if demanded at any time during the stage up until the window time expires

Note 2:- Condition of Termination

0 - Terminates at end of stage

1 - Terminates when associated phase gains right of way

2 - Terminates when associated phase loses right of way

3 - Terminates at end of minimum green

4 - Terminates at end of maximum green

5 - Terminates subject to special conditioning

Note 3: - Phase Types

T - Traffic

F - Filter Arrow

PD- Pedestrian

I - Indicative Arrow

PU- Puffin

D - Dummy

TN - Toucan Nearside

L - L.R.T

TF - Toucan Farside

S - Switched Sign

C - Cycle

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[illegible]

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[illegible][illegible][illegible]

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C = CONFLICTING & OPPOSING

0 = OPPOSING ONLY

. = NO CONFLICT

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STARTING INTERGREEN PERIOD	12
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PROHIBITED / ALTERNATIVE STAGE MOVEMENTS UP TO 16 STAGES

[illegible]

TICK AS REQUIRED		
MODE	These Restrictions are for	No Restrictions
Urban Traffic Control		✓
Cableless Linking		✓
MOVA		✓
Hurry Call		
Manual		✓
Priority		
Emergency		

NOTE:

✓	= Permitted
P	= Prohibited
X	= Ignore
N'	= ALTERNATIVE MOVES

where 'N' is number of stage to move to

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DETECTOR AND PUSH BUTTONS - LOCATIONS AND FUNCTIONS

ID	Location / Title	Type	Demand	Extend	Non - Latching	Call Delay	Cancel Delay	Extension	Paired Detector	Kerbside Extension	DFM Group	Error State	Special Instructions
1	AIN1	V		A,DA				4.2			1	Y	MOVA Det 1
2	AX2	V	A	A,DA				4.2			1	A	MOVA Det 2
3	BIN3	V		B,DA				4.2			1	Y	MOVA Det 3
4	BX4	V	B	B,DA				4.2			1	A	MOVA Det 4
5	PBU C - Pole 1	PB	C		✓						3	Y	
6	PBU C - Pole 4	PB	C		✓						3	Y	
7	Kerbside - Pole 1	K	C						5/6	1.0	2	I	
8	On Crossing - Pole 4	O		C				0.5			1	A	
9	PBU C - Pole 2	PB	C		✓						3	Y	
10	PBU C - Pole 3	PB	C		✓						3	Y	
11	Kerbside - Pole 3	K	C						9/10	1.0	2	I	
12	On Crossing - Pole 2	O		C				0.5			1	A	
13	Tactile fault - Pole 2										4	A	
14	Tactile fault - Pole 4										4	A	

☒	Tick appropriate boxes	
Group	Detector Fault Monitoring Times (DFM)	
	Fail Active after (mins)	Fail Inactive after (hours)
1	30	18
2	10	24
3	10	0
4	10	120

Detector Types:-			Error States
V Vehicle	C Clearance	SU Speed Upstream	A Fail Active
PB Pushbutton	Q Queue	SD Speed Downstream	I Fail Inactive
K Kerbside	U1 Uni Directional 1		Y Fail in input state
O On-Crossing	U2 Uni Directional 2		
Paired Detector:-			
Specify ID of paired Uni-directional or kerbside/pushbuttons			

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MOVA DETECTORS								
No.	Name / Description	Action	No.	Name / Description	Action	No.	Name / Description	Action
1	AIN1		25	DS1	MOVA Dataset 1	49		
2	AX2		26	DS2	MOVA Dataset 2	50		
3	BIN3		27	DS3	MOVA Dataset 3	51		
4	BX4		28	DS4	MOVA Dataset 4	52		
5			29	SF1	UTC Hold A	53		
6			30	SF2	Priority A	54		
7			31	SF3	UTC Hold B	55		
8			32	SF4	Priority B	56		
9	PBU C	Demands C	33	Bus Trigger Call A	Calls Bus Priority on A	57		
10			34	Bus Trigger Hold A	Holds Bus Priority on A	58		
11			35	Bus Trigger Cancel	Cancels Bus Priority on A	59		
12			36	Bus Trigger Call B	Calls Bus Priority on B	60		
13			37	Bus Trigger Hold B	Holds Bus Priority on B	61		
14			38	Bus Trigger Cancel	Cancels Bus Priority on B	62		
15			39			63		
16			40			64		
17			41	Kerbside	Green man extension	65		
18			42			66		
19			43			67		
20			44			68		
21			45			69		
22			46			70		
23			47			71		
24			48			72		

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ADDITIONAL SWITCHES & LED'S (ON MANUAL PANEL)		
POSITION	USE	MOMENTARY *
SWITCH 1		
LED 1	MOVA Enabled (Flashing) UTC Mode (Solid)	
SWITCH 2		
LED 2		
SWITCH 3		
LED 3		
HURRY CALL AUX 4		
HIGHER PRIORITY AUX 5		

HURRY CALL(S)					
HC(0) ALWAYS HAS PRIORITY OVER HC(1)					
HURRY CALL No	CALL STAGE	CALL DELAY	CALL HOLD	CALL PREVENT	NOTES
CALL LIMIT	LOW				
	HIGH				

OPTION FILTER GREEN ARROW IN MANUAL			
TO APPEAR	NOT TO APPEAR	DEMAND DEPENDANT	INSERT TICK

PART TIME MODE	
SWITCH OFF IN STAGE(S)	

*

NORMAL ACTION IS A TOGGLE ACTION
(PRESS ONCE FOR ACTIVE, PRESS AGAIN TO MAKE INACTIVE)

PUT TICK IF MOMENTARY ACTION IS REQUIRED
(ACTIVE ONLY WHILE BUTTON IS PRESSED)

IF SWITCH IS MOMENTARY OR NOT USED THEN THE ASSOCIATED LED IS AVAILABLE FOR OTHER PURPOSES

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MODE PRIORITY

ASSIGN NUMBER 1 = TOP PRIORITY
2 = SECONDY PRIORITY etc.
X = NOT USED

_____ PART TIME
_____ EMERGENCY VEHICLE
_____ HURRY CALL(S)
1 _____ URBAN TRAFFIC CONTROL
2 _____ SELECTED MANUAL CONTROL (1)
3 _____ SELECTED FIXED TIME OR VA (2)
5 _____ CABLELESS LINKING
4 _____ MOVA
6 _____ EITHER VEHICLE ACTUATED
_____ OR FIXED TIME

PERMANENTLY DISABLED MODES

MODES TO BE DISABLED	STREAM			
	0	1	2	3
PART TIME (3)				
EMERGENCY VEHICLE				
HURRY CALL(S)				
SELECTED MANUAL CONTROL (3)				
URBAN TRAFFIC CONTROL				
SELECTED FIXED TIME OR VA				
CABLELESS LINKING				
BUS PRIORITY				
VEHICLE ACTUATED				
FIXED TIME				
✓ TICK IF REQUIRED TO BE DISABLED.				

MANUAL FACILITY ENABLED

ALWAYS	✓
WHEN HANDSET PLUGGED IN	
WHEN "MND" COMMAND ENTERED	

FACILITY SWITCH POSITION	NOT OPERATIVE INSERT (X)
VA	
FIXED TIME	
CLF	

(3) THESE MODES MAY ONLY BE DELETED ON ALL MODES SIMULTANEOUSLY

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FIXED CYCLE TIME MODE STAGE SEQUENCE								
Phase	A	B	C	D	E	F	G	H
Stage	1st	2nd	3rd	4th	5th	6th	7th	8th
Time								

OR

ALTERNATIVELY FIXED CYCLE RUNNING TO CURRENT TIMETABLED MAXIMA																
Phase	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Demand Dependent			✓													
Phase	Q	R	S	T	U	V	Q	W	X	Y	Z					
Demand Dependent																

MANUAL SELECTION		
Button No.	Stage(s) called	Street Names - If Required
ALL RED		All Red Stage(s)
1	1	Traffic Stages
2	2	Toucan Stage

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UTC ONLY - MANUAL MODE, NO LAMP SUPPLY, LAMP FAILURES etc., CONFIRM					
TICK AS APPROPRIATE:-	G1/G2	MC	DF	LF1	LF2
MANUAL MODE SELECTED AND OPERATIVE	✓	✓			
NO LAMP POWER and/or SIGNALS OFF	✓				
DETECTOR FAULT			✓		
SELECT SWITCH NOT IN NORMAL POSITION		✓			
ANY LAMP FAILURE				✓	
2nd RED LAMP FAILURE ON ANY PHASE				✓	✓

Leeds UTC Lamp Monitoring Specification:

On any lamp failure the LF1 reply bit shall be returned.

On the 1st red lamp failure to any phase conflicting with a pedestrian phase, the LF1 reply bit shall be returned and the detector fault monitor lamp shall flash.

On any second red lamp failure to any phase LF2 shall be returned.

On any second red lamp failure to any phase conflicting with a pedestrian phase, the LF2 reply bit shall be sent and the controller shall switch off.

On any lamp failure there shall be no intergreen extensions.

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Telent (Microsense Type) Timetable Data

Timetable Data			
No.	Day Type	Time	Event List No.
1	WKD	07:00:00	2
2	WND	07:00:00	1
3	WKD	09:30:00	1
4	FRI	15:00:00	3
5	MON-THU	15:45:00	3
6	WEK	19:00:00	4
7	WEK	07:00:00	9
8	WEK	22:00:00	10
9	WEK	00:00:00	12
10	WEK	00:00:01	14
11			
12			
13			
14			
15			
16			

Day Types:-

MON - Monday
TUE - Tuesday
WED - Wednesday
THU - Thursday
FRI - Friday
SAT - Saturday
SUN - Sunday
WKD - Mon - Fri
WND - Sat - Sun
WEK - Every day

XMO - All except Mon
XTU - All except Tue
XWE - All except Wed
XTH - All except Thu
XFR - All except Fri
XSA - All except Sat
XSU - All except Sun

Event List Data								
Event List No.	Event Action 1		Event Action 2		Event Action 3		Event Action 4	
	Type	Params	Type	Params	Type	Params	Type	Params
1	TTS	1						
2	TTS	2						
3	TTS	3						
4	TTS	4						
5	TTS	5						
6	TTS	6						
7	TTS	7						
8	TTS	8						
9	TEF	1=ON	Enable Bleepers					
10	TEF	1=OFF	Disable Bleepers					
11	TEF	2=ON	Disable PTM					
12	TEF	2=OFF	Enable PTM					

Event Types:-

TTS - timetable phase timing set
TCF - timetable CLF
TEF - timetable event flag
TPT - timetable part-time
TSF - timetable switched sign
TDO - timetable detector override
TDT - timetable detector timing set

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IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

EXT C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED "-" - BOOLEAN "AND" "+" - BOOLEAN "OR"

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS	
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Mode during which conditions apply :- VEHICLE ACTUATED / FIXED TIME

STREAM 1

[illegible]

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

EXT C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED “.” - BOOLEAN “AND” “+” - BOOLEAN “OR”

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